

LABORATORIUM CLOUD COMPUTING & BIG DATA

PROGRAM STUDI INFORMATIKA /

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BAGIAN 1 — Langkah-Langkah Instalasi Hadoop

Ikuti setiap langkah di bawah ini dan sertakan screenshot sesuai ketentuan. Tempel screenshot pada area yang telah disediakan dan isi keterangan singkat.

Langkah 1 — Update & Upgrade System Packages

Jalankan perintah berikut untuk memastikan sistem Ubuntu 24.04 sudah up-to-date:

SS-1

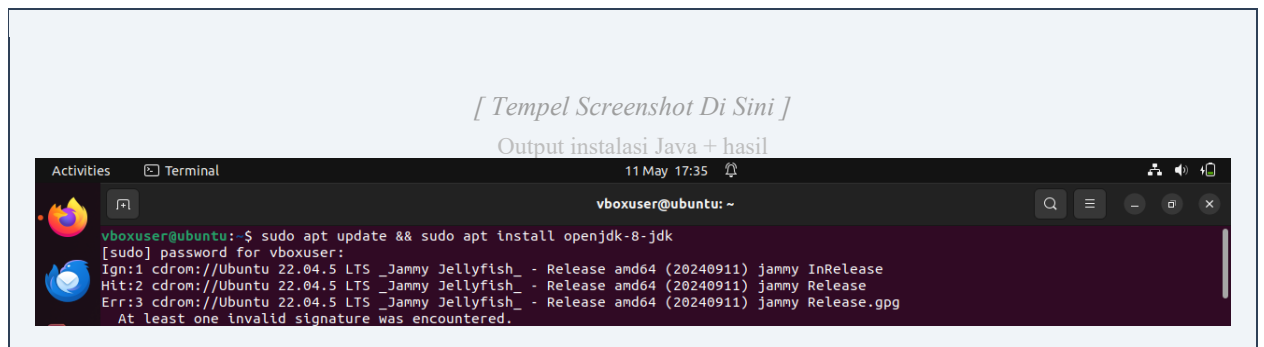


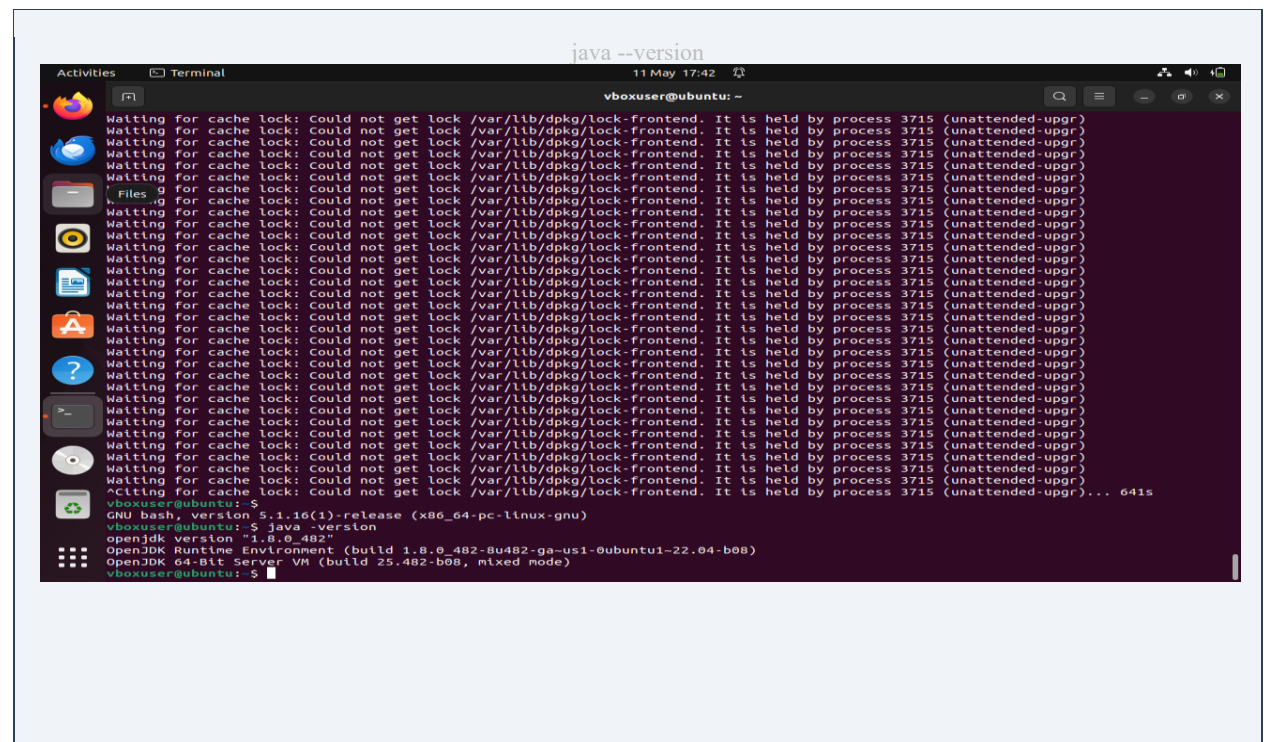
Gambar – SS-1

Langkah 2 — Instalasi Java (OpenJDK)

Hadoop memerlukan Java. Install OpenJDK dan verifikasi versinya:

SS-2



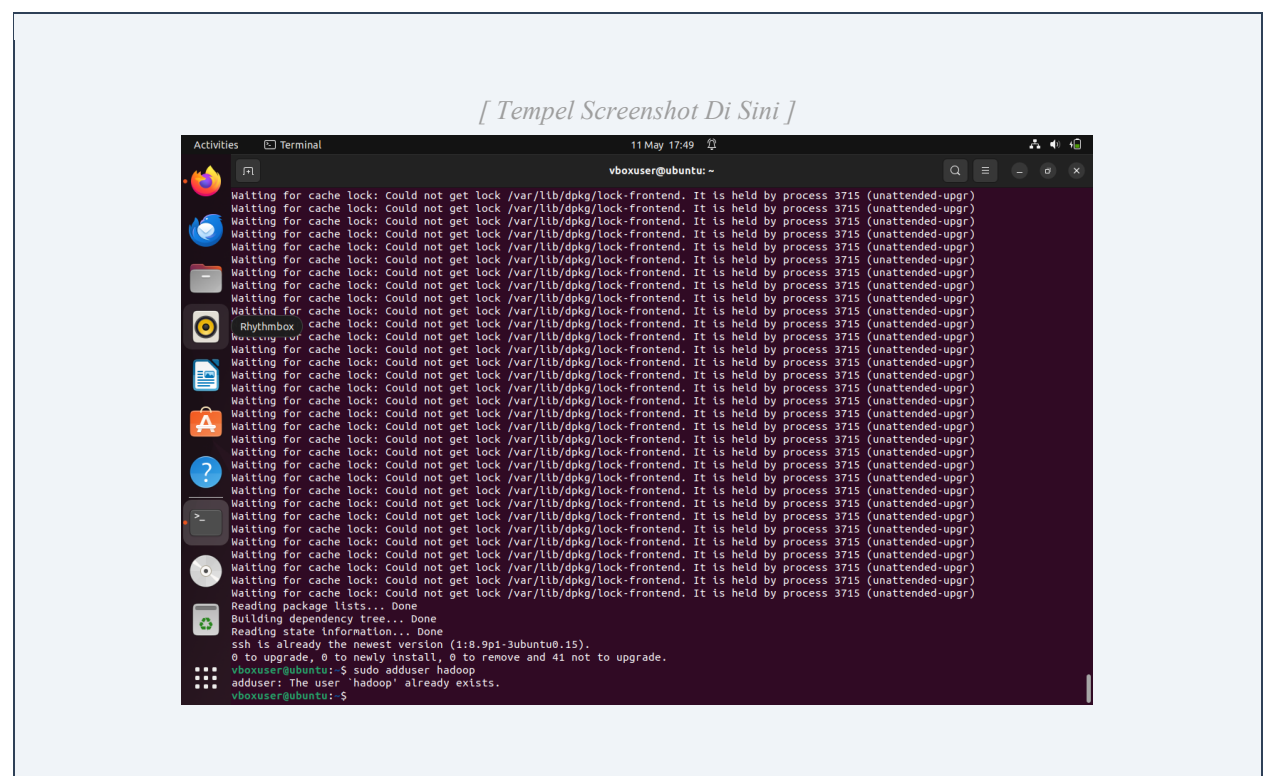


Gambar – SS-2

Langkah 3 — Membuat User Hadoop

Buat user khusus untuk Hadoop dan tambahkan ke grup sudo:

SS-3



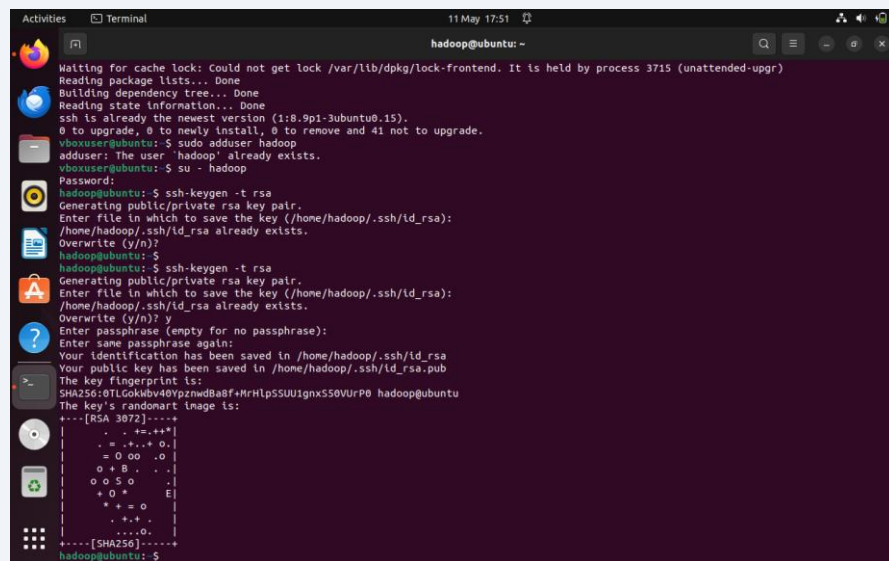
Output perintah adduser dan usermod

Gambar – SS-3

Langkah 4 — Konfigurasi SSH Passwordless

Generate SSH key agar node Hadoop dapat berkomunikasi tanpa password:

SS-4



[Tempel Screenshot Di Sini]

Output ssh-keygen (tampilkan fingerprint)

Gambar – SS-4

Langkah 5 — Download & Ekstrak Apache Hadoop

Download binary Hadoop dari server Apache dan ekstrak:

SS-5

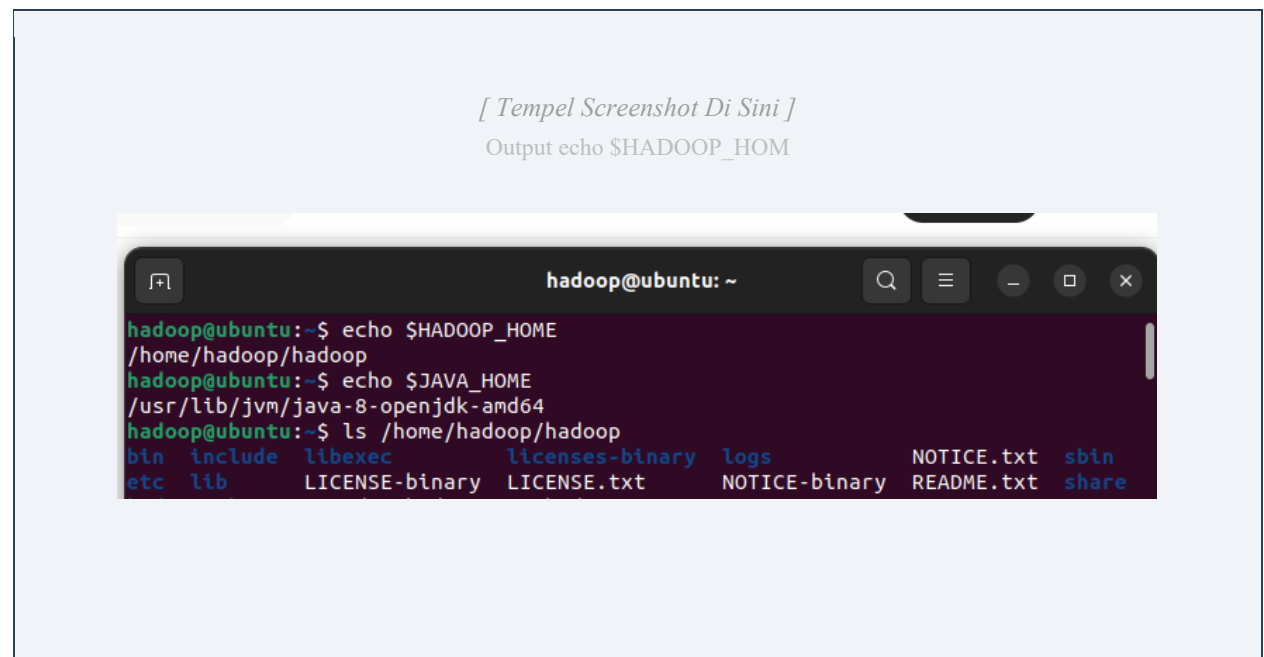


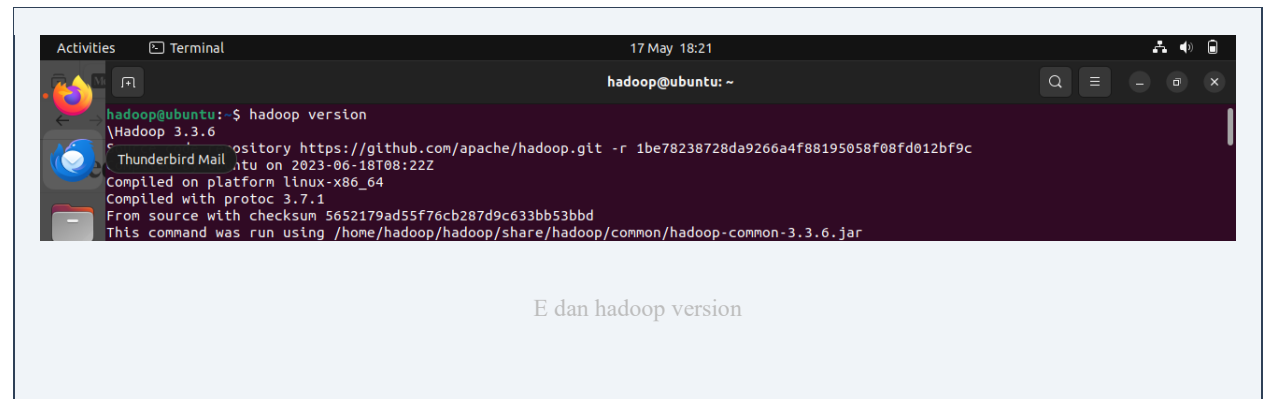
Gambar – SS-5

Langkah 6 — Konfigurasi Environment Variable

Set environment variable Hadoop di ~/.bashrc dan verifikasi:

SS-6



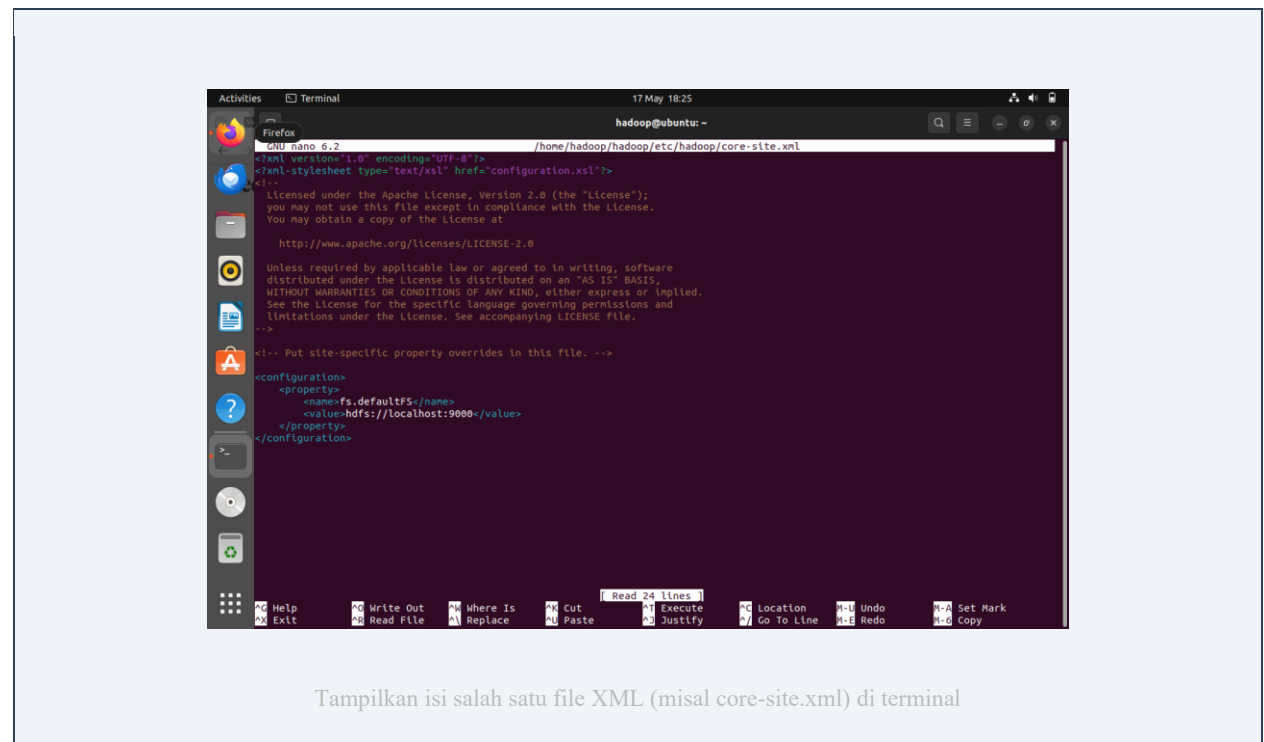


Gambar – SS-6

Langkah 7 — Konfigurasi File XML Hadoop

Edit file konfigurasi XML (core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml):

SS-7



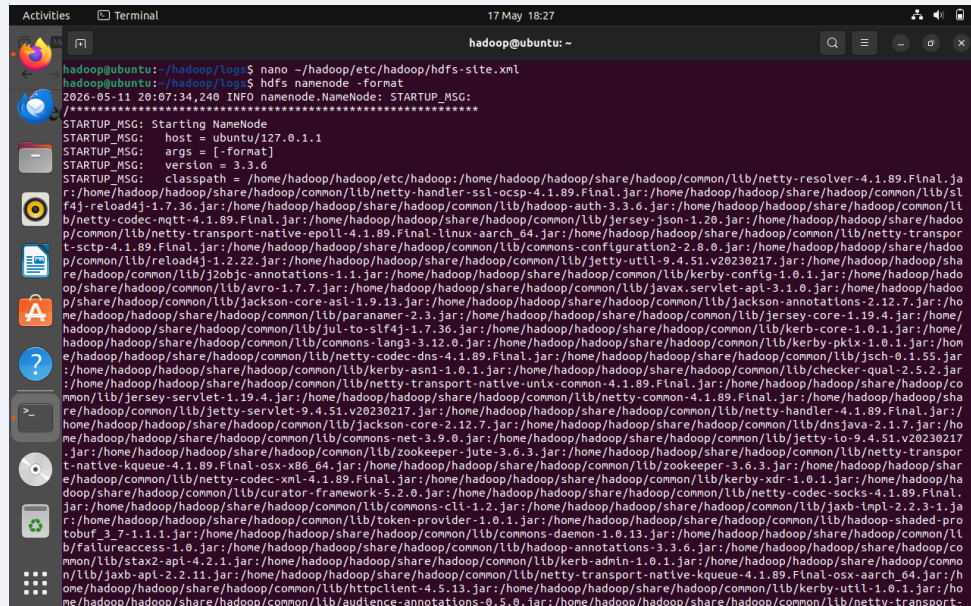
Gambar – SS-7

Langkah 8 — Format NameNode HDFS

Format NameNode untuk pertama kali sebelum menjalankan Hadoop:

- `hdfs namenode -format`

SS-8



```

hadoop@ubuntu:~/hadoop/logs$ nano ~/hadoop/etc/hadoop/hdfs-site.xml
hadoop@ubuntu:~/hadoop/logs$ hdfs namenode -format
2020-05-11 20:07:34,240 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = ubuntu/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.3.6
STARTUP_MSG: classpath = /home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/common/lib/netty-resolver-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/ssl-faj-reload4j-1.7.36.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-auth-3.3.6.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-nat-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jackson-annotations-2.12.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-epoll-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/javax-servlet-api-3.1.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-configuration2-2.8.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/reload4j-1.2.22.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-util-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jzobjc-annotations-1.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-config-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/avro-1.7.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/javax-servlet-api-3.1.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/paranamer-2.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-core-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-lang3-3.12.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-pkix-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-dns-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-asn1-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/checker-qual-2.5.2.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-unix-common-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jersey-servlet-1.19.4.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-common-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-servlet-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-handler-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jackson-core-2.12.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/dnsjava-2.1.7.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-net-3.9.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jetty-io-9.4.51.v20230217.jar:/home/hadoop/hadoop/share/hadoop/common/lib/zookeeper-jute-3.6.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-kqueue-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/zookeeper-3.6.3.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-mx-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerby-xdr-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/curator-framework-5.2.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-codec-socks-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-cli-1.2.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/token-provider-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-shaded-proto-3.7-1.1.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/commons-daemon-1.0.13.jar:/home/hadoop/hadoop/share/hadoop/common/lib/failureaccess-1.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/hadoop-annotations-3.3.6.jar:/home/hadoop/hadoop/share/hadoop/common/lib/stax2-api-4.2.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/kerb-adint-1.0.1.jar:/home/hadoop/hadoop/share/hadoop/common/lib/jaxb-aot-2.2.11.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-native-kqueue-4.1.89.Final.jar:/home/hadoop/hadoop/share/hadoop/common/lib/audience-annotations-0.5.0.jar:/home/hadoop/hadoop/share/hadoop/common/lib/netty-transport-

```

Output hdfs namenode -format (bagian 'successfully formatted')

Gambar – SS-8

Langkah 9 — Start DFS & YARN + Verifikasi JPS

Jalankan layanan Hadoop dan verifikasi prosesnya:

SS-9



```

hadoop@ubuntu:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ubuntu]
Starting resourcemanager
Starting nodemanagers
hadoop@ubuntu:~$ jps
53620 ResourceManager
53222 DataNode
54070 Jps
53414 SecondaryNameNode
53096 NameNode
53740 NodeManager
hadoop@ubuntu:~$

```

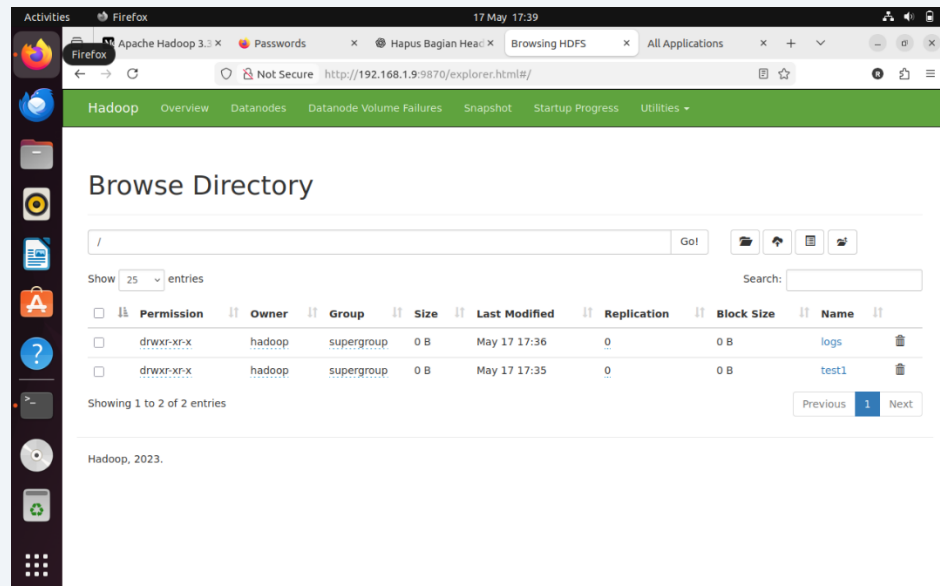
Output start-dfs.sh, start-yarn.sh, dan hasil perintah jps

Gambar – SS-9

Langkah 10 — Akses Web UI Hadoop (Namenode)

Buka browser dan akses Hadoop NameNode Web UI:

SS-10



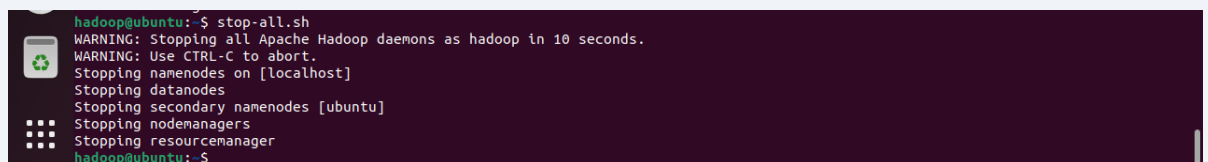
Tampilan browser localhost:9870 (Hadoop NameNode Web UI)

Gambar – SS-10

Langkah 11 — Stop DFS & YARN + Verifikasi JPS

Hentikan layanan Hadoop setelah selesai digunakan:

SS-11



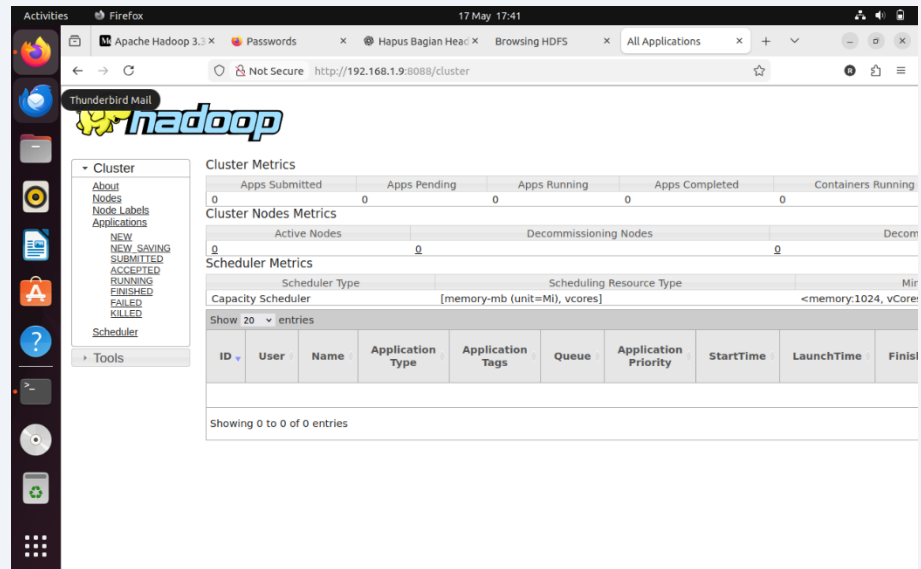
Output stop-dfs.sh, stop-yarn.sh, dan hasil jps setelah dihentikan

Gambar – SS-11

Langkah 12 (Opsional) — Akses YARN Resource Manager

Akses YARN Resource Manager Web UI (opsional, poin bonus):

SS-12 (Optional)



Tampilan browser localhost:8088 (YARN Resource Manager)

Gambar – SS-12 (Optional)

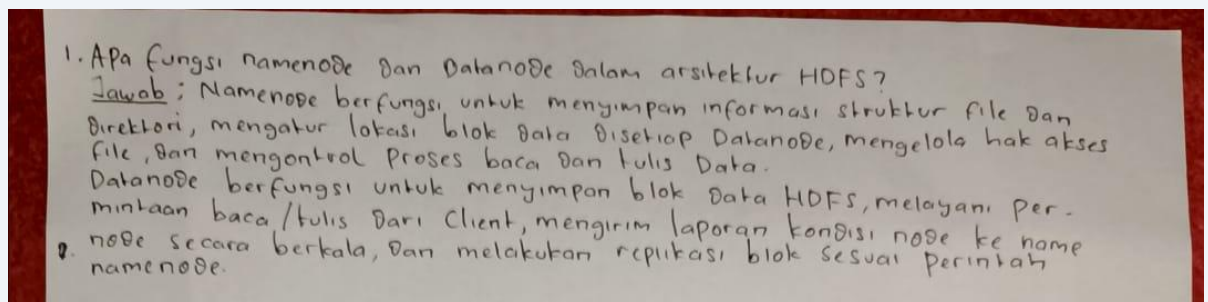
BAGIAN 2 — Pertanyaan & Jawaban Teori

PERHATIAN: Bagian 2 ini dikerjakan dan dijawab dengan **TULIS TANGAN** di kertas, kemudian difoto dan dilampirkan di bawah ini.

1. Apa fungsi NameNode dan DataNode dalam arsitektur HDFS?

→ Lampirkan foto jawaban tulis tangan di bawah ini:

Foto Jawaban No. 1



[Tempel Screenshot Di Sini]

Foto lembar jawaban tulis tangan — pertanyaan 1

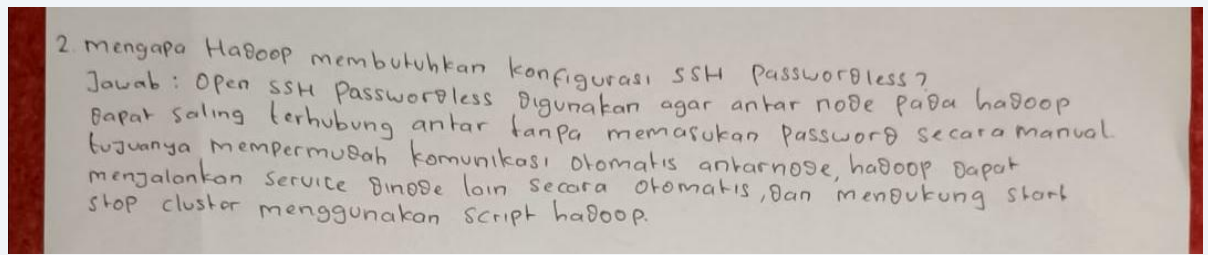
Gambar – Foto Jawaban No. 1

2. Mengapa Hadoop membutuhkan konfigurasi SSH passwordless?

→ Lampirkan foto jawaban tulis tangan di bawah ini:

Foto Jawaban No. 2

[Te



[Tempel Screenshot Di Sini]

Foto lembar jawaban tulis tangan — pertanyaan 2

Gambar – Foto Jawaban No. 2

3. Apa perbedaan pseudo-distributed mode dengan fully distributed mode?

→ Lampirkan foto jawaban tulis tangan di bawah ini:

Foto Jawaban No. 3

[Tempel Screenshot Di Sini]

Foto lembar jaw

3. Perbedaan pseudo-distributed mode dan fully distributed mode.		
Jawab :		
<u>aspek</u>	<u>Pseudo-distributed mode</u>	<u>Fully-distributed mode</u>
- Jumlah komputer	- 1 komputer	- Banyak komputer
- Fungsi node	- Semua service berjalan di satu mesin	- Tiap service bisa di mesin beda
- Tujuan	- Testing dan pembelajaran	- produksi nyata
- konfigurasi	- lebih sederhana	- lebih kompleks
- Performa	- terbatas	- lebih cepat dan scalable
- Jaringan	- localhost	- Jaringan antar server

aban tulis tangan — pertanyaan 3

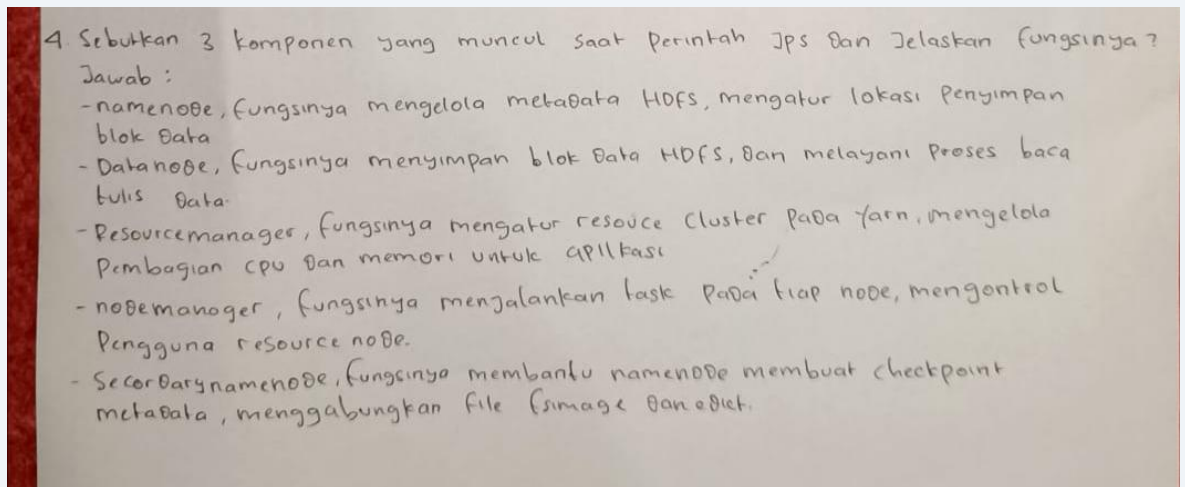
Gambar – Foto Jawaban No. 3

4. Sebutkan minimal 3 komponen yang muncul saat perintah jps dan jelaskan fungsinya!

→ Lampirkan foto jawaban tulis tangan di bawah ini:

Foto Jawaban No. 4

[Temp



el Screenshot Di Sini]

Foto lembar jawaban tulis tangan — pertanyaan 4

Gambar – Foto Jawaban No. 4